SCRFD-2.5G
Face Detection Performance Report

DeepStream + TensorRT FP16 · GPU Blur Pipeline

Work Item INTER-69 · Face Detection + GPU Blur

MEASURED REPORT INTER-69-SCRFD-2.5G · October 2026

Report ID	INTER-69-SCRFD-2.5G
Work Item	INTER-69 · Face Detection + GPU Blur
Model	SCRFD-2.5G
Task	Face detection + GPU blur
Device	NVIDIA GeForce RTX 5080 (16 GB)
Runtime	DeepStream + TensorRT (FP16)
Input size	640 × 640
Script	scrfd_face_detection.py
Status	Measured · performance evidence recorded

Prepared in the layout and control style of the xis.ai AI Departmental Handbook v1.0.

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1 Executive Summary

Purpose. This report records measured performance for **SCRFD-2.5G** in the INTER-69 face detection + GPU blur pipeline on NVIDIA GeForce RTX 5080.

Measured finding. SCRFD-2.5G is the fastest model in this work item at 398.5 FPS on a 1920×1080 video file, with the lowest GPU utilisation (18.3%) and the lowest per-stage latency. Total detections: 42,802.

1.1 Common settings

Setting	Value
Device	NVIDIA GeForce RTX 5080 (16 GB)
Runtime	DeepStream + TensorRT (FP16)
Model	SCRFD-2.5G
Detection size	640 × 640
Confidence / NMS	0.35 / 0.4
Frames	1800 (warmup 30 excluded from timings)
Frames dropped	0
Bottleneck	encode (NVENC)

1.2 Summary metrics (video file, 1920×1080)

Metric	Value
Avg Pipeline FPS	398.5
Estimated Max FPS	983.70
Headroom	1.73×
Total Detections	42,802
Avg Faces / Frame	23.78
Max Faces / Frame	35

1.3 Latency per frame (ms)

Stage	mean	max
Detect (nvinfer / TensorRT)	0.42	3.09
Post (decode + NMS)	0.62	2.19
Blur (CUDA kernels)	0.15	0.60
Encode (NVENC)	1.02	2.43

Note: Detect and Blur rows highlighted in green. Total Detections highlighted in yellow.

1.4 Resource usage

Metric	Value
CPU cores used (avg)	1.77
GPU utilisation avg / peak	18.3% / 23%
GPU memory peak	4158 MB

2 Module Execution Placement

Module	Device
Stream Decode (NVDEC)	GPU

Face Detection Model (TensorRT)	GPU
Post-processing (decode + NMS)	CPU (lightweight)
Face Blur (Custom CUDA kernels)	GPU
Video Encode (NVENC)	GPU
Stream Mux / Demux / Queue (DeepStream)	GPU

Almost the entire pipeline runs on GPU. Only the tiny detector output tensor decoding + NMS runs on CPU.

3 High-Rate RTSP Test

A local high-rate RTSP source (FFmpeg without `-re`) was used to measure maximum pipeline throughput.

Metric	Value
Avg Pipeline FPS	408.50
Source FPS	428.60
Frames Processed	14,332
Frames Dropped	6
Headroom	2.58×
Estimated Max FPS	1052.89

Latency per frame (ms) — high-rate RTSP

Stage	mean	max
Detect (nvinfer / TensorRT)	0.58	2.53
Post (decode + NMS)	0.78	3.94
Blur (CUDA kernels)	0.17	2.06
Encode (NVENC)	0.77	5.42

Resource usage — high-rate RTSP

Metric	Value
CPU cores used (avg)	2.17
GPU utilisation avg / peak	15.03% / 21%
GPU memory peak	4050 MB

4 Key Takeaways

• **SCRFD-2.5G** achieved **398.5 FPS** on a 1920×1080 video file with 0 frames dropped.

• Total detections: **42,802** (avg 23.78 faces/frame, max 35).

• GPU utilisation avg / peak: **18.3% / 23%**; memory peak: **4158 MB**.

• Real live streams (cameras / Rutube) are limited by source rate (~25–30 FPS). The pipeline itself is much faster.

• High-rate RTSP test reached **408.50 FPS** with headroom 2.58×.